

```

fit_data_path = 'e:\SHARE\Fe\Data\analysis\06_fit_with_J0\Sym4D_cutsAndFits';
%fit_data_file = 'FitEn_cut200_1DsbG_dirGH_dE20_fix_bg_slope.mat'; % fit 1D
background with no Q-slope
fit_data_file =
'FitEn_cut200ref200_2D_constFFCut_dirGH_dE20_fix_bg_slope.mat';%Externaly 2-
exponent fit fixed bg at high momentum.
%fit_data_file = 'FitEn_cut200ref110_2D_dirGH_dE20_fix_bg_slope.mat';%Externaly
2-exponent fit fixed bg at high momentum.

if ~exist('fit_resEi200','var')
    ld = load(fullfile(fit_data_path,fit_data_file));
    if isfield(ld,'fit_resEi200')
        fit_resEi200 = ld.fit_resEi200;
    else
        fit_resEi200 = ld.fit_res;
    end
end
data_path = 'e:\SHARE\Fe\Data\analysis\06_fit_with_J0\sym4D_cutsAndFits';
fit_file = 'multicuts_fit_dataGH_ei200meV.mat';
if ~exist('cuts2fit200','var')
    ld = load(fullfile(data_path,fit_file));
    cuts2fit200 = ld.cuts2fit200;
end
cuts2fit = cuts2fit200;
fit_res = fit_resEi200;
mi = maps_instrument(200,600,'S');
sample=IX_sample(true,[1,0,0],[0,1,0],'cuboid',[0.04,0.03,0.02]);
%cl = {cuts2fit200.cut110_sel};
cl = {cuts2fit200.cut200_sel};
%cl = {cut(cuts2fit200.cut200_sel,[0,0.02,1],[])};
for i=1:numel(cl)
    cl{i} = cl{i}.set_instrument(mi);
    cl{i} = cl{i}.set_sample(sample);

    %cuts2fit.cutsS_list{i} = cuts2fit.cutsS_list{i}.set_instrument(mi);
    %cuts2fit.cutsS_list{i} = cuts2fit.cutsS_list{i}.set_sample(sample);
end

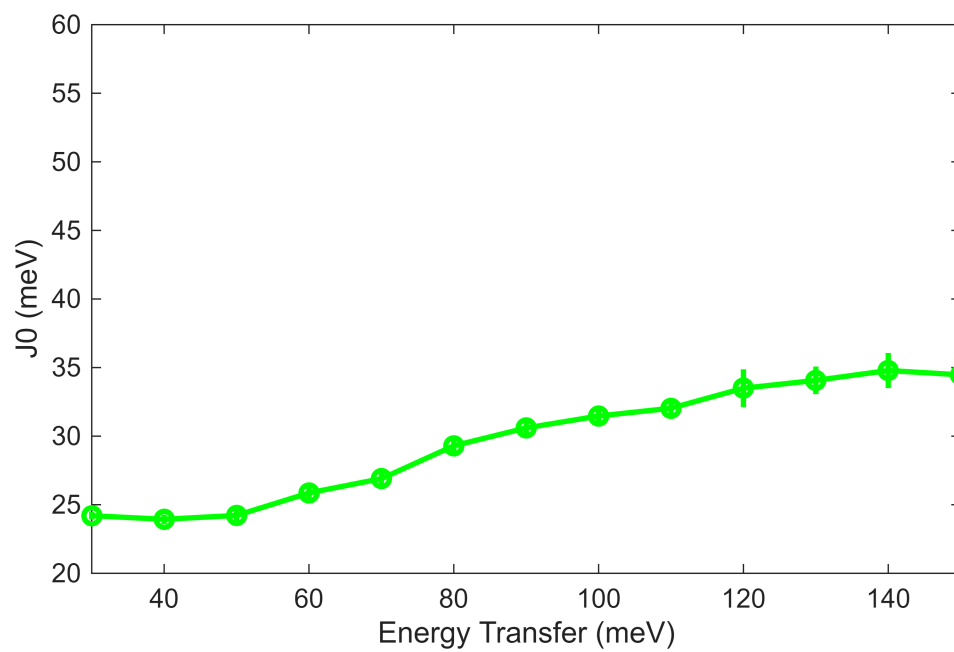
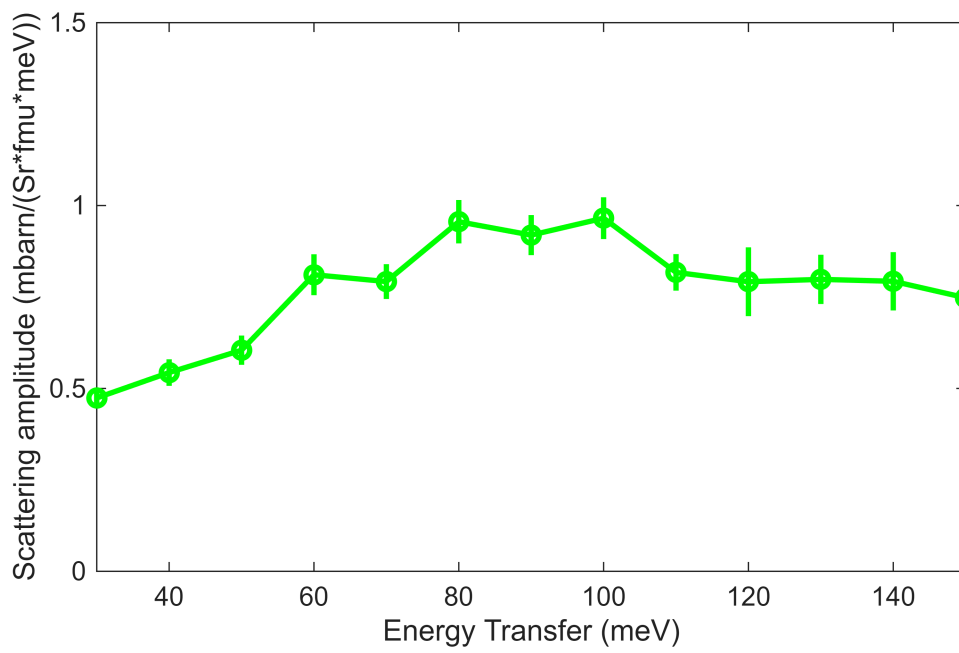
the2Dcuts = cl;

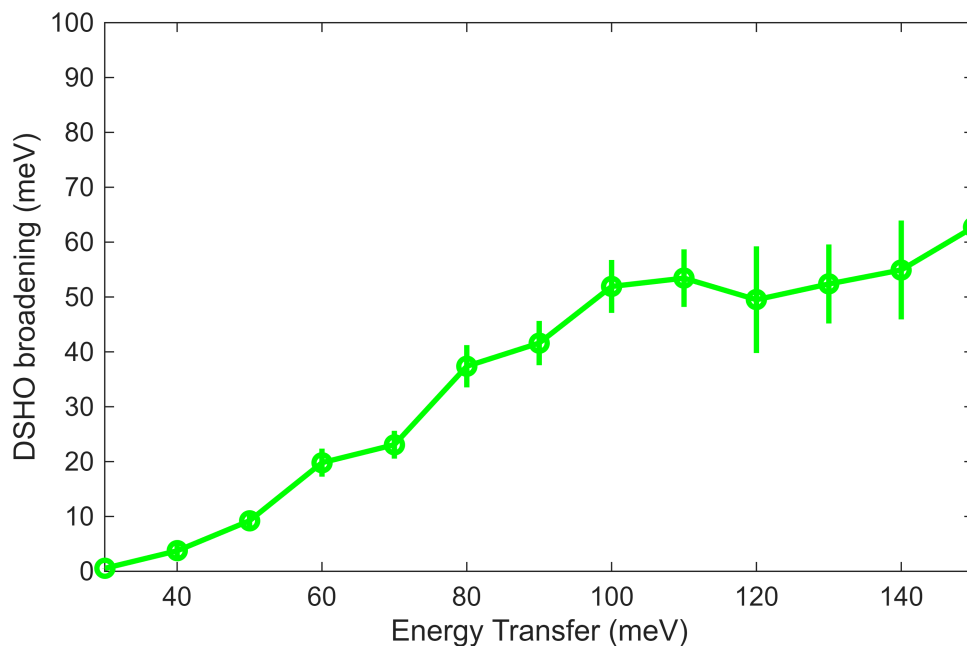
```

```

[Sgn1,J0,G] = plot_SJG_results(fit_res,'g',{ 'S','J0','gamma'},{[0,1.5],[20,60],
[0,100]});

```





```
dE = min(Sgn1.x(2:end)-Sgn1.x(1:end-1));
e_min = Sgn1.x(1);
```

```
En = 100
```

```
En =
100
```

```
dE_step = 2; %original energy transfer step data were binned to. No point in
going finer
half_dE = 10;
en_idx = floor((En-e_min)/dE)+1;
%the2Dcuts = cuts2fit.cutsS_list;%(1:2);
%the2Dcuts = {cut(cuts2fit.cut110_sel,[0,0.02,2],[])};%(1:2);

%init_bg_params =
%cellfun(@(par)[par(1),par(3)],init_bg_params,'UniformOutput',false); % bg
%in case of 2-D fitting
%
[fit_obj,fit_par,figa,figb]=fit_single_set(the2Dcuts,En,half_dE,dE_step,init_fg_p
arams,init_bg_params,false)
lfit_par = fit_res.all_fit_par(en_idx);
en_range = lfit_par.en_range;
[fit_obj,fit_par,figa,figb]=refit_singleset_peaks(the2Dcuts,en_range,lfit_par,0.2
,true,false);
```

Warning: Dataset:75 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 6.7507

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested
 *** Step 1 of 1; Read data for 499694 pixels -- processing data... -----> included 449627 pixels

```
-----
Number of points in input object: 750621
Fraction of object selected: 66.57% (= 499694 points) ; Access speed: 2.08GB/sec
Fraction of object retained: 59.90% (= 449627 points)
```

```

**** Total time in cut(sqw)   :      0.0sec Including:
      Data access time fraction: 49.8%; Transf. time frac.: 48.6%; Resulting pix preparation:  0.0%
-----
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for 171420 pixels -- processing data... -----> retained 160535 pixels
-----
Number of points in input object: 781729
      Fraction of object selected: 21.93% (= 171420 points) ; Access speed: 1.96GB/sec
      Fraction of object retained: 20.54% (= 160535 points)

**** Total time in cut(sqw)   :      0.0sec Including:
      Data access time fraction: 27.2%; Transf. time frac.: 47.4%; Resulting pix preparation: 25.3%
-----
*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested
*** Step 1 of 1; Read data for 269494 pixels -- processing data... -----> included 239599 pixels
-----
Number of points in input object: 750621
      Fraction of object selected: 35.90% (= 269494 points) ; Access speed: 2.03GB/sec
      Fraction of object retained: 31.92% (= 239599 points)

**** Total time in cut(sqw)   :      0.0sec Including:
      Data access time fraction: 37.0%; Transf. time frac.: 55.9%; Resulting pix preparation:  0.0%
-----
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for 84698 pixels -- processing data... -----> retained 79735 pixels
-----
Number of points in input object: 781729
      Fraction of object selected: 10.83% (= 84698 points) ; Access speed: 2.05GB/sec
      Fraction of object retained: 10.20% (= 79735 points)

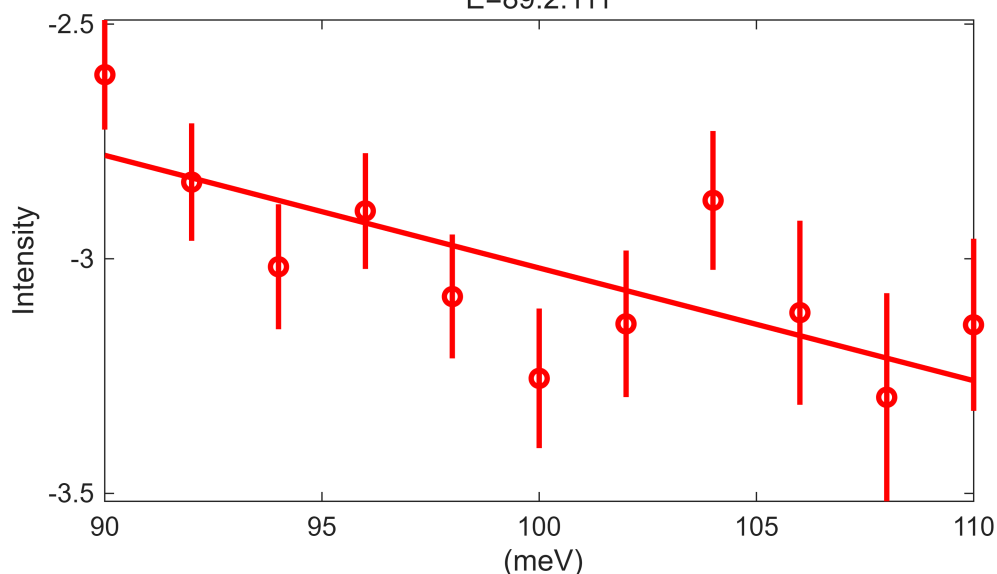
**** Total time in cut(sqw)   :      0.0sec Including:
      Data access time fraction: 17.3%; Transf. time frac.: 63.2%; Resulting pix preparation: 19.5%
-----
*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested
*** Step 1 of 1; Read data for 160535 pixels -- processing data... -----> included 160535 pixels
-----
Number of points in input object: 160535
      Fraction of object selected: 100.00% (= 160535 points) ; Access speed: 1.71GB/sec
      Fraction of object retained: 100.00% (= 160535 points)

**** Total time in cut(sqw)   :      0.0sec Including:
      Data access time fraction: 57.7%; Transf. time frac.: 36.5%; Resulting pix preparation:  0.0%
-----
*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested
*** Step 1 of 1; Read data for 79735 pixels -- processing data... -----> included 79735 pixels
-----
Number of points in input object: 79735
      Fraction of object selected: 100.00% (= 79735 points) ; Access speed: 2.01GB/sec
      Fraction of object retained: 100.00% (= 79735 points)

**** Total time in cut(sqw)   :      0.0sec Including:
      Data access time fraction: 21.9%; Transf. time frac.: 74.3%; Resulting pix preparation:  0.0%
-----

```

Fe_ei200_align_sym3.sqw
 $\delta \leq \zeta \leq 1.86$ in $[2, \zeta, 0]$, $-0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2$
 $E=89:2:111$



Warning: Dataset 2:12 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 7.2727

Beginning fit (max 20 iterations)-----

Starting point

Total time = 2.375s Reduced Chi^2 = 10.984

Free parameter values:

51.92 0.9655 31.47 0.4236 0.5387

Iteration = 1

Total time = 11.7044s Reduced Chi^2 = 1.1161 Levenberg-Marquardt = 0.1

Free parameter values:

44.24 0.6043 30.57 0.1628 0.1033

Iteration = 2

Total time = 21.18s Reduced Chi^2 = 1.0894 Levenberg-Marquardt = 0.01

Free parameter values:

39.84 0.6536 29.88 0.1347 0.05379

Iteration = 3

Total time = 30.6194s Reduced Chi^2 = 1.0876 Levenberg-Marquardt = 0.001

Free parameter values:

42.76 0.6903 30.21 0.1273 0.04219

Iteration = 4

Total time = 39.965s Reduced Chi^2 = 1.0887 Levenberg-Marquardt = 0.0001

Free parameter values:

42.41 0.6855 30.17 0.1283 0.04363

Total time = 42.411s Reduced Chi^2 = 1.09 Levenberg-Marquardt = 0.001

Free parameter values:

42.42 0.6858 30.17 0.1283 0.04355

Total time = 44.8266s Reduced Chi^2 = 1.0898 Levenberg-Marquardt = 0.01

Free parameter values:

42.51 0.6872 30.18 0.128 0.0431

Total time = 47.2777s Reduced Chi^2 = 1.0893 Levenberg-Marquardt = 1

Free parameter values:

42.71 0.6901 30.2 0.1274 0.04227

Total time = 49.6729s Reduced Chi^2 = 1.0883 Levenberg-Marquardt = 100

Free parameter values:

42.75 0.6903 30.21 0.1273 0.0422

Total time = 52.0084s Reduced Chi^2 = 1.0906 Levenberg-Marquardt = 10000

Free parameter values:

42.76	0.6903	30.21	0.1273	0.04219
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*** No improvement in chi-squared over previous iteration ***

Total time = 52.0162s Reduced Chi^2 = 1.0876

Free parameter values:

42.76	0.6903	30.21	0.1273	0.04219
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Fit converged:

Parameter values (free parameters with error estimates):

Foreground parameter values:

1	1		
2	8		
3	42.76	+/-	6.579
4	0.6903	+/-	0.08529
5	0		
6	30.21	+/-	0.8189
7	0		
8	0		
9	0		
10	0		

Background parameter values:

Function 1:

1	0.1273	+/-	0.0167
2	-0.01231		
3	0		

Function 2:

1	0.04219	+/-	0.008692
2	-0.02401		
3	0		

Covariance matrix for free parameters:

1.0000	0.9432	0.8461	-0.7538	-0.9437
0.9432	1.0000	0.8294	-0.8619	-0.8643
0.8461	0.8294	1.0000	-0.6383	-0.8544
-0.7538	-0.8619	-0.6383	1.0000	0.5269
-0.9437	-0.8643	-0.8544	0.5269	1.0000

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested

*** Step 1 of 1; Read data for 781729 pixels -- processing data... -----> included 710069 pixels

Number of points in input object: 781729

Fraction of object selected: 100.00% (= 781729 points) ; Access speed: 2.09GB/sec
Fraction of object retained: 90.83% (= 710069 points)

**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 54.9%; Transf. time frac.: 44.0%; Resulting pix preparation: 0.0%

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested

*** Step 1 of 1; Read data for 160535 pixels -- processing data... -----> included 146588 pixels

Number of points in input object: 160535

Fraction of object selected: 100.00% (= 160535 points) ; Access speed: 1.88GB/sec
Fraction of object retained: 91.31% (= 146588 points)

**** Total time in cut(sqw) : 0.0sec Including:

Data access time fraction: 44.7%; Transf. time frac.: 51.2%; Resulting pix preparation: 0.0%

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested
 *** Step 1 of 1; Read data for 75753 pixels -- processing data... -----> included 69276 pixels

Number of points in input object: 75753

Fraction of object selected: 100.00% (= 75753 points) ; Access speed: 1.49GB/sec
 Fraction of object retained: 91.45% (= 69276 points)

**** Total time in cut(sqw) : 0.0sec Including:
 Data access time fraction: 26.2%; Transf. time frac.: 69.7%; Resulting pix preparation: 0.0%

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
 *** Step 1 of 1; Read data for 781729 pixels -- processing data... -----> retained 710069 pixels

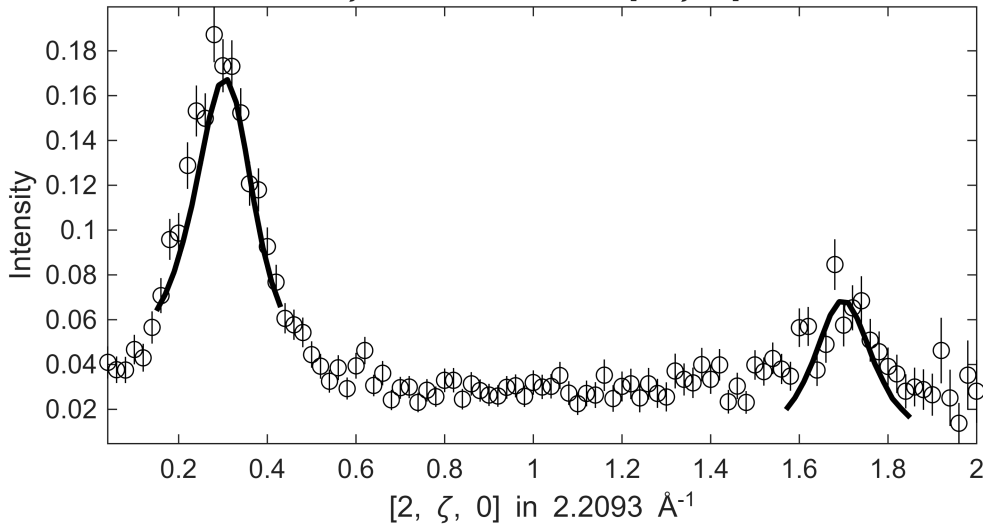
Number of points in input object: 781729

Fe_ei200_align_sym3.sqw

S=0.69±0.085; J0= 30±0.82; gamma=42.7565±6.57887

-0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 90 ≤ E ≤ 110

ζ=-0.01:0.02:2.01 in [2, ζ, 0]



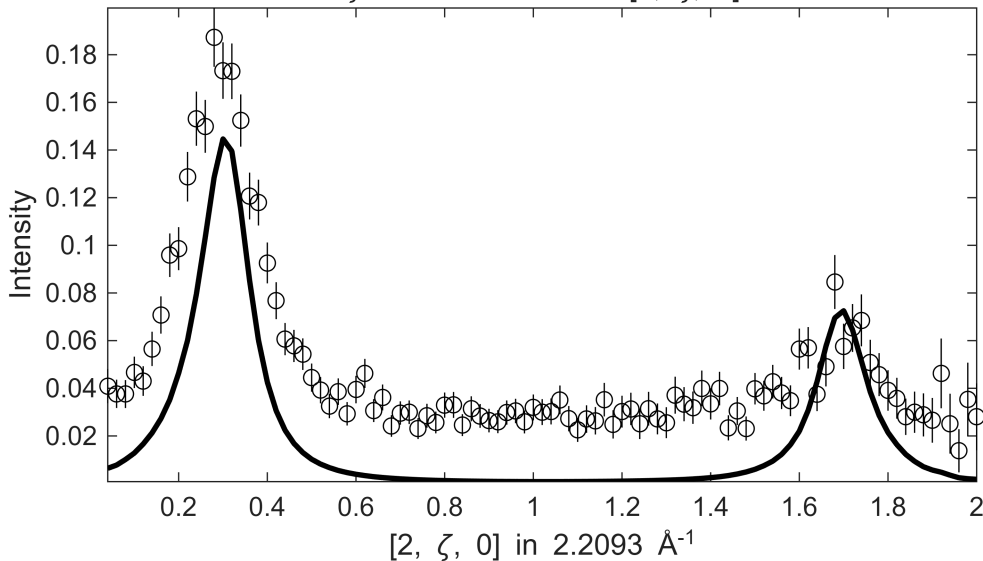
Fraction of object selected: 100.00% (= 781729 points) ; Access speed: 2.08GB/sec
 Fraction of object retained: 90.83% (= 710069 points)

**** Total time in cut(sqw) : 0.1sec Including:
 Data access time fraction: 27.2%; Transf. time frac.: 45.4%; Resulting pix preparation: 27.4%

Fe_ei200_align_sym3.sqw

-0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, η] , 90 ≤ E ≤ 110

ζ=-0.01:0.02:2.01 in [2, ζ, 0]

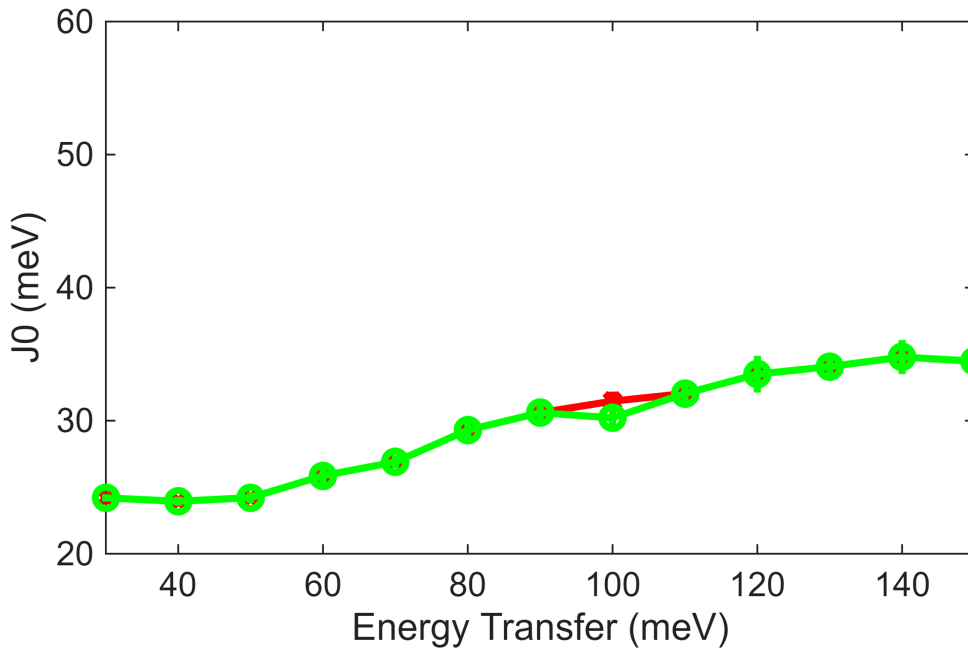
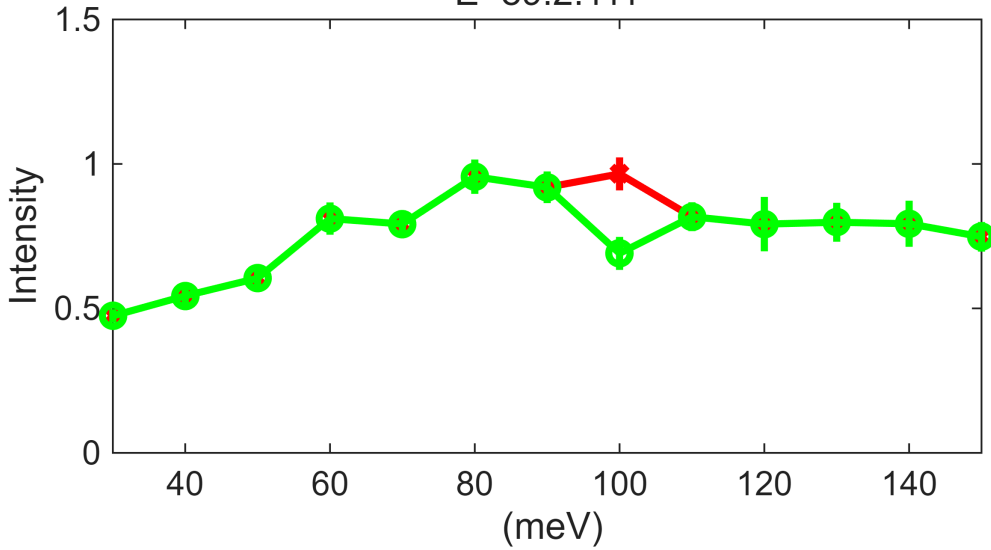


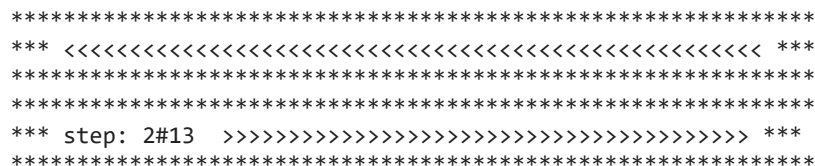
```

%fit_res.all_fit_par(en_idx) = fit_par;
%fit_res.all_fit_par(en_idx).p = fit_par.p;
Sgn1.signal(en_idx) = fit_par.p(4);
J0.signal(en_idx) = fit_par.p(6);
G.signal(en_idx) = fit_par.p(3);
plot_SJG_results(fit_res, 'g', {'S', 'J0', 'gamma'}, {[0,1.5],[20,60],[0,100]} ,
{Sgn1,J0,G});

```

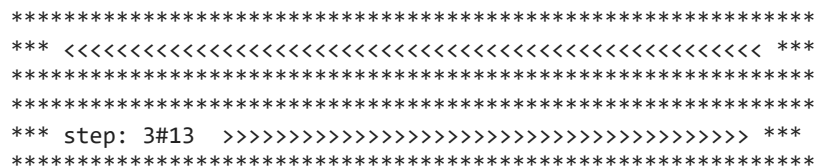
Fe_ei200_align_sym3.sqw
1.86 in $[2, \zeta, 0]$, $-0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq$
E=89:2:111



$$\xi \in [-0.1, 0.1] \text{ in } [2-\xi, 0, \bar{0}] , -0.1 \leq \eta \leq 0.1 \text{ in } [2, 0, \eta] , 20 \leq \zeta \leq 20.02 \text{ in } [2, \zeta, 0]$$


S=0.41±0.025; J0= 24±0.36; gamma=2.13491±0.1069
 $\leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $30 \leq E$
 $\zeta=-0.01:0.02:2.01$ in $[2, \zeta, 0]$

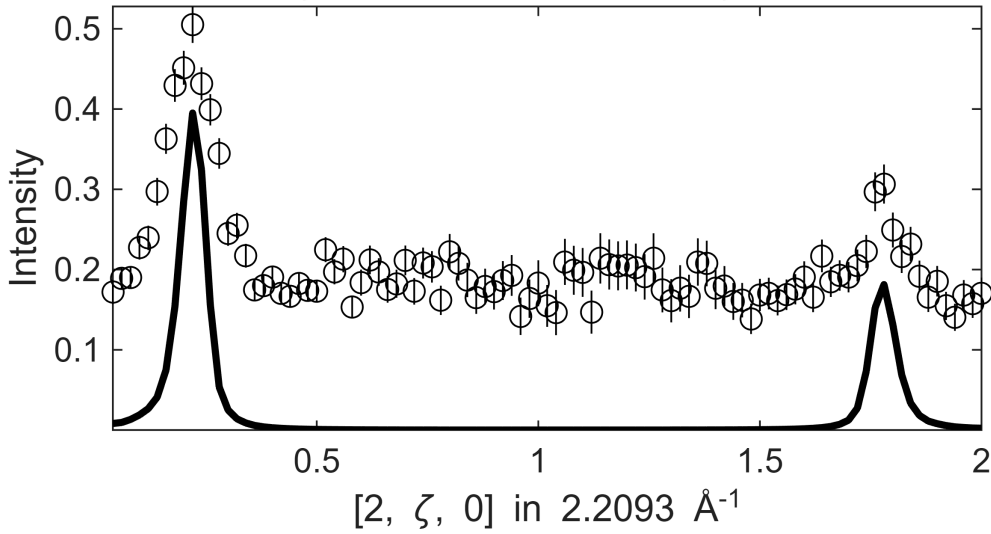


$$\xi \in [-0.1, 0.1] \text{ in } [2-\xi, 0, \bar{0}] , -0.1 \leq \eta \leq 0.1 \text{ in } [2, 0, \eta] , 30 \leq \zeta \leq 300 \text{ in } [2, \zeta, 0]$$


S=0.46±0.032; J0= 24±0.37; gamma=6.68393±0.357662
 $\leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $40 \leq E$
 $\zeta=-0.01:0.02:2.01$ in $[2, \zeta, 0]$

Fe_ei200_align_sym3.sqw

$\leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $40 \leq \epsilon$
 $\zeta=-0.01:0.02:2.01$ in $[2, \zeta, 0]$



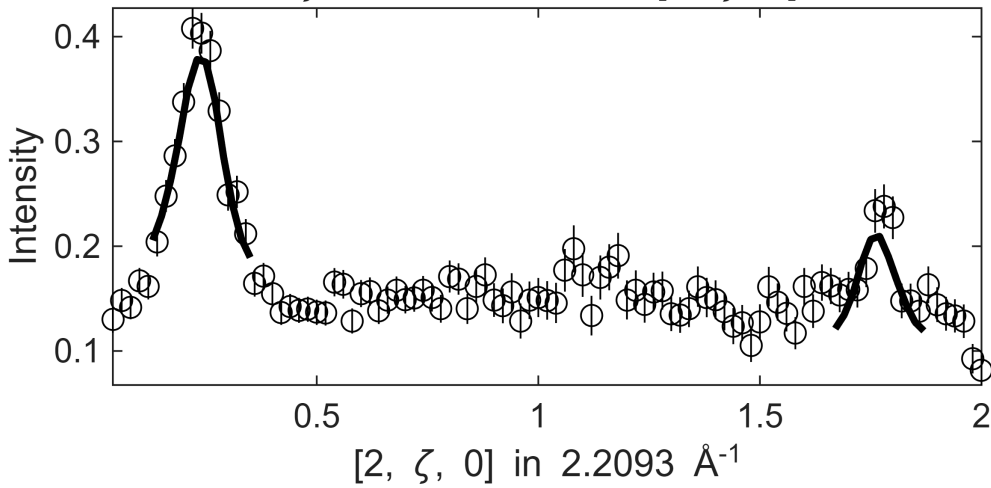
```
*****
*** <<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<< ***
*****
*****
*** step: 4#13 >>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>> ***
*****
```

Warning: Dataset:3 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.27003

Fe_ei200_align_sym3.sqw

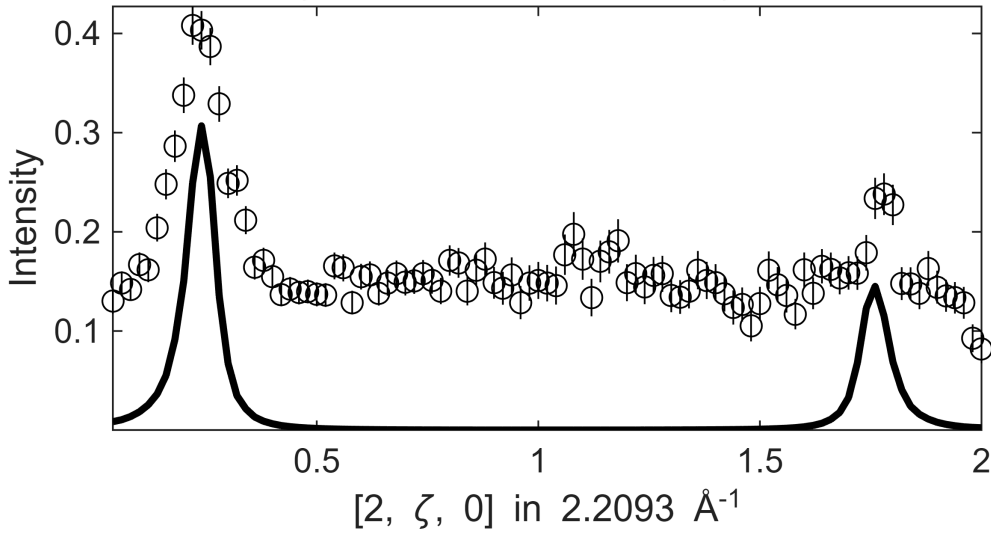
S=0.52±0.11; J0= 26±0.46; gamma=12.2458±3.93624

$\leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $50 \leq \epsilon$
 $\zeta=-0.01:0.02:2.01$ in $[2, \zeta, 0]$



Fe_ei200_align_sym3.sqw

$\leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $50 \leq \epsilon$
 $\zeta=-0.01:0.02:2.01$ in $[2, \zeta, 0]$



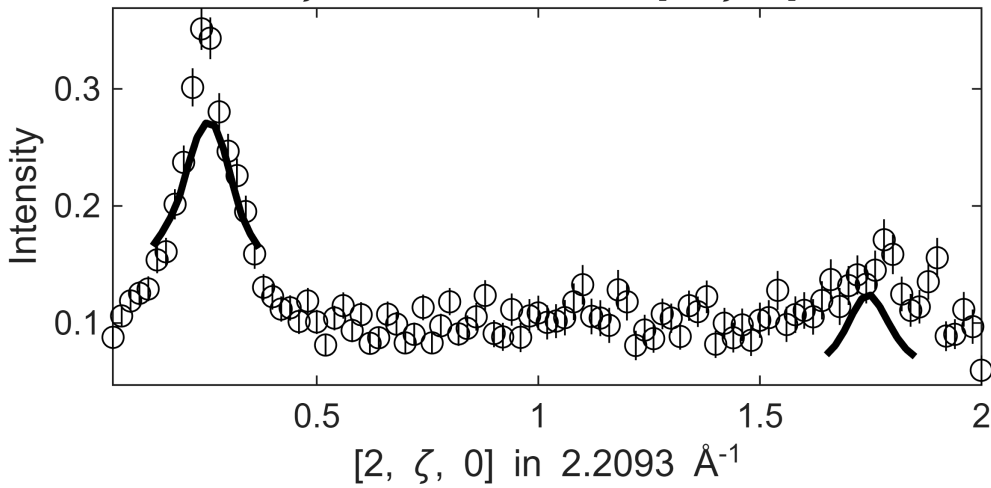
```
*****
*** <<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<< ***
*****
*****
*** step: 5#13 >>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>> ***
*****
```

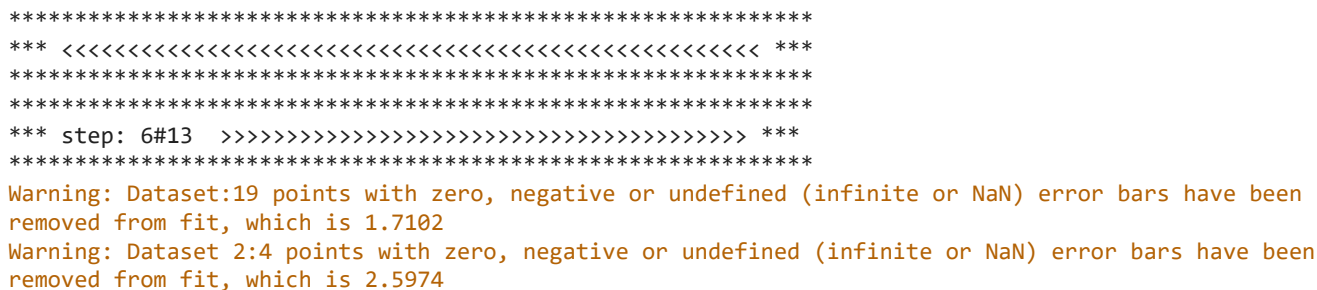
Warning: Dataset:7 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.63006

Fe_ei200_align_sym3.sqw

S=0.39±0.055; J0= 27± 0.8; gamma=19.4467±1.66322

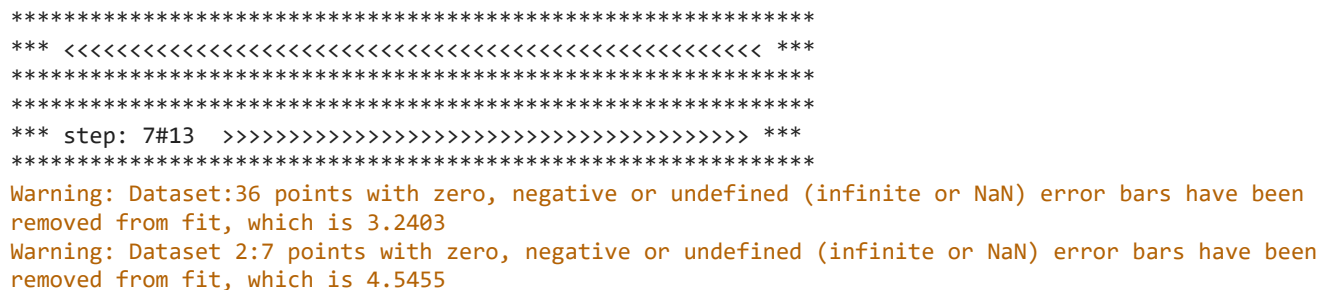
$\leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $60 \leq \epsilon$
 $\zeta=-0.01:0.02:2.01$ in $[2, \zeta, 0]$



$$\xi \in [-0.1, 0.1] \text{ in } [2-\xi, 0, 0], -0.1 \leq \eta \leq 0.1 \text{ in } [2, 0, \eta], 60 \leq \zeta \leq 120 \text{ in } [2, \zeta, 0]$$


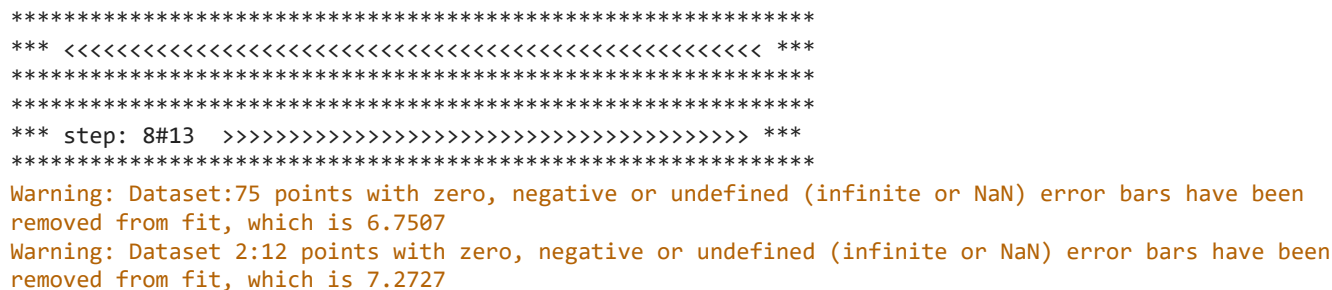
S=0.61±0.16; J0= 29± 1; gamma=33.0106±7.81184
 $\leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $70 \leq \zeta$ in $[2, \zeta, 0]$



$$\xi \in [-0.1, 0.1] \text{ in } [2-\xi, 0, \bar{0}] \text{ , } \eta \in [-0.1, 0.1] \text{ in } [2, 0, \eta] \text{ , } \zeta \in [-0.01:0.02:2.01] \text{ in } [2, \zeta, 0]$$

$$S=0.54\pm 0.12; J_0=29\pm 0.82; \gamma=29.8419\pm 7.67238$$

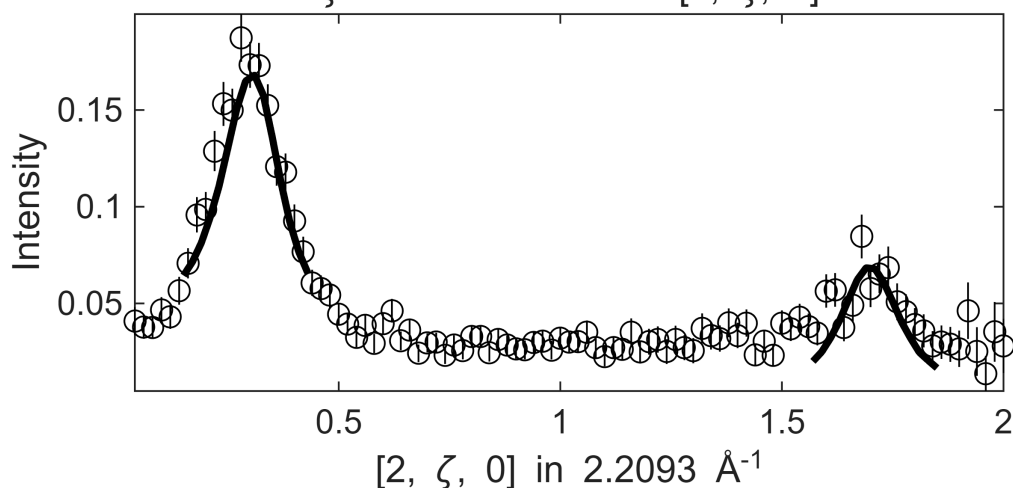
$$\leq \xi \leq 0.1 \text{ in } [2-\xi, 0, 0], -0.1 \leq \eta \leq 0.1 \text{ in } [2, 0, \eta], 80 \leq E$$

$$\zeta=-0.01:0.02:2.01 \text{ in } [2, \zeta, 0]$$


$$\xi \in [-0.1, 0.1] \text{ in } [2-\xi, 0, 0], -0.1 \leq \eta \leq 0.1 \text{ in } [2, 0, \eta], 80 \leq E$$


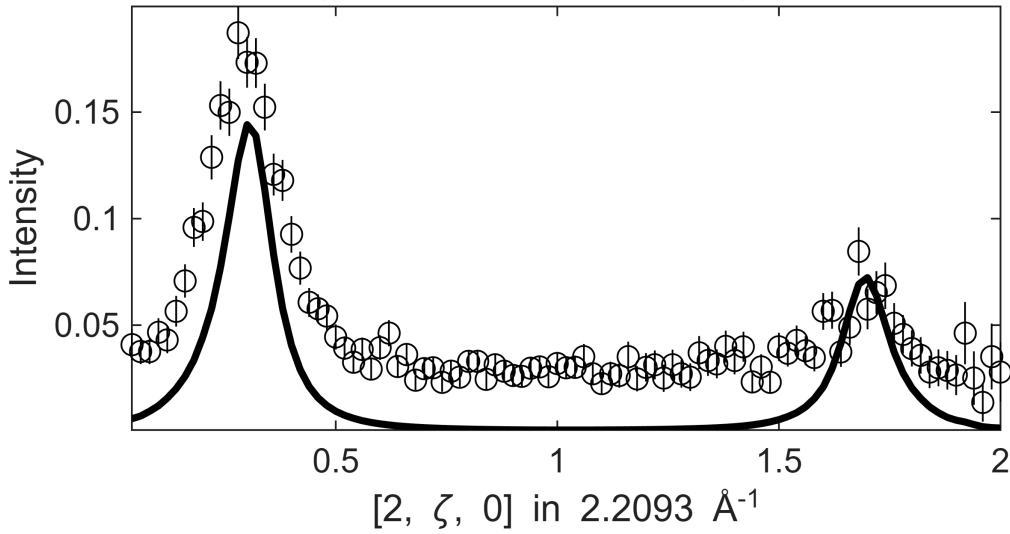
$S=0.66\pm0.15$; $J_0=30\pm 1$; $\gamma=40.8822\pm9.5371$

$$\leq \xi \leq 0.1 \text{ in } [2-\xi, 0, 0], -0.1 \leq \eta \leq 0.1 \text{ in } [2, 0, \eta], 90 \leq E$$

$$\zeta = -0.01:0.02:2.01 \text{ in } [2, \zeta, 0]$$


Fe_ei200_align_sym3.sqw

$\xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $90 \leq E$
 $\zeta = -0.01:0.02:2.01$ in $[2, \zeta, 0]$



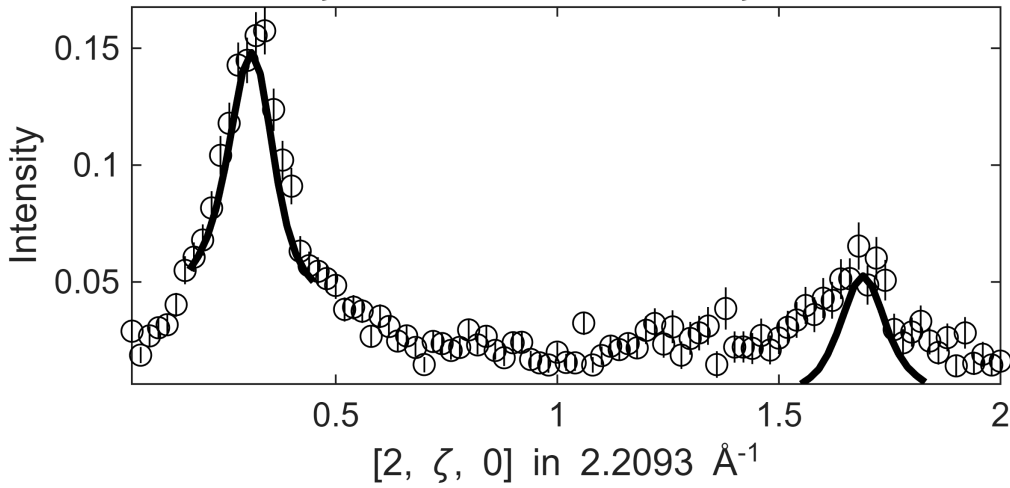
```
*****
*** <<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<< ***
*****
*****
*** step: 9#13 >>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>> ***
*****
```

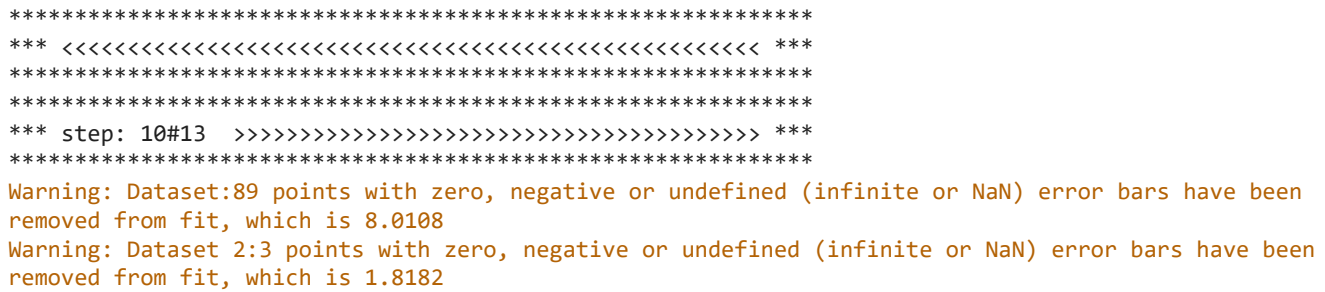
Warning: Dataset:108 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 9.721
Warning: Dataset 2:10 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 6.0606

Fe_ei200_align_sym3.sqw

S=0.45±0.034; J0= 31±0.52; gamma=31.0411±3.34091

$\xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $100 \leq E$
 $\zeta = -0.01:0.02:2.01$ in $[2, \zeta, 0]$



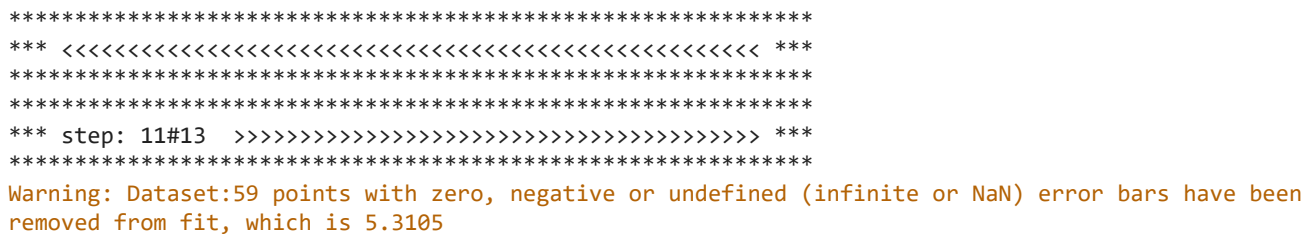
$$\xi \leq 0.1 \text{ in } [2-\xi, 0, 0], -0.1 \leq \eta \leq 0.1 \text{ in } [2, 0, \eta], 100 \leq \zeta \leq 1000 \text{ in } [2, \zeta, 0]$$


$S=0.51\pm0.03$; $J_0=33\pm0.48$; $\gamma=42.153\pm3.14042$

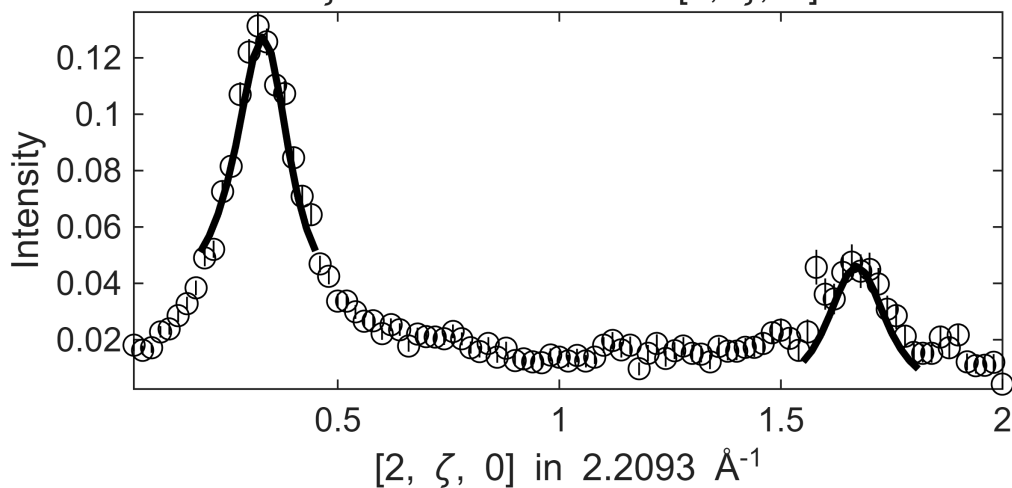
$$\xi \in [-0.1, 0.1] \text{ in } [2-\xi, 0, 0], -0.1 \leq \eta \leq 0.1 \text{ in } [2, 0, \eta], 110 \leq E$$

$$\zeta = -0.01:0.02:2.01 \text{ in } [2, \zeta, 0]$$


$$\xi \leq 0.1 \text{ in } [2-\xi, 0, 0], -0.1 \leq \eta \leq 0.1 \text{ in } [2, 0, \eta], 110 \leq E$$

$$\zeta = -0.01:0.02:2.01 \text{ in } [2, \zeta, 0]$$


$S=0.54\pm0.082$; $J_0=34\pm0.59$; $\gamma=48.2114\pm6.94968$

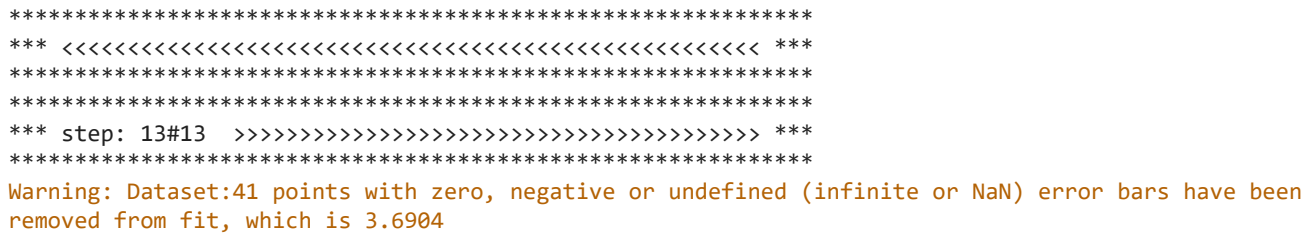
$$\xi \leq 0.1 \text{ in } [2-\xi, 0, 0], -0.1 \leq \eta \leq 0.1 \text{ in } [2, 0, \eta], 120 \leq \zeta \leq 180 \text{ in } [2, \zeta, 0]$$


$$\xi \leq 0.1 \text{ in } [2-\xi, 0, 0], -0.1 \leq \eta \leq 0.1 \text{ in } [2, 0, \eta], 120 \leq \epsilon$$

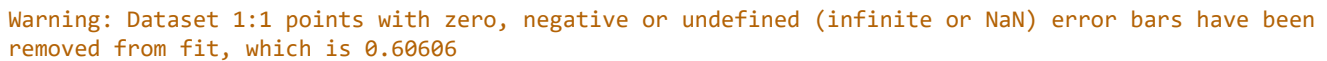
$$\zeta = -0.01:0.02:2.01 \text{ in } [2, \zeta, 0]$$


Fe_ei200_align_sym3.sqw

$$\xi \leq 0.1 \text{ in } [2-\xi, 0, 0], -0.1 \leq \eta \leq 0.1 \text{ in } [2, 0, \eta], 130 \leq \zeta \leq 150 \text{ in } [2, \zeta, 0]$$

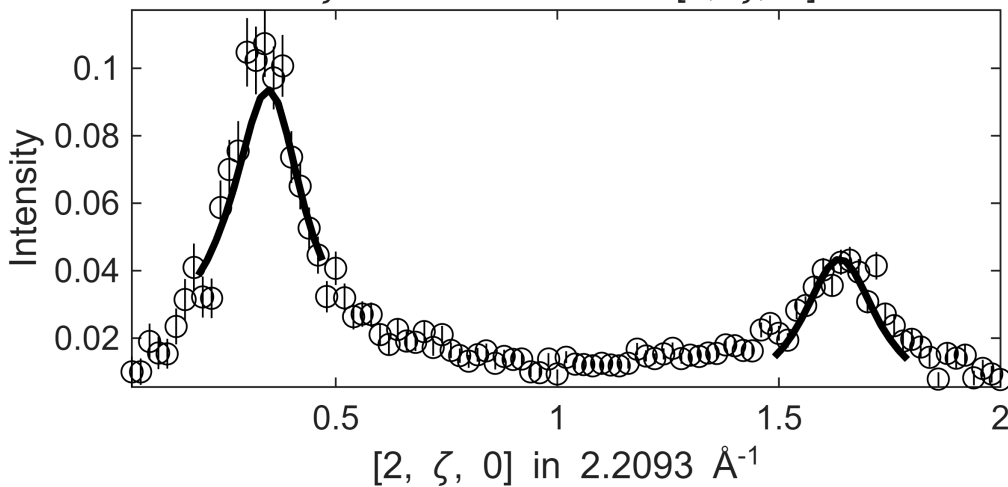

$$\xi \leq 0.1 \text{ in } [2-\xi, 0, 0], -0.1 \leq \eta \leq 0.1 \text{ in } [2, 0, \eta], 130 \leq \zeta \leq 150 \text{ in } [2, \zeta, 0]$$


: 1.8 in [2, ζ, 0] , -0.1 ≤ ξ ≤ 0.1 in [2-ξ, 0, 0] , -0.1 ≤ η ≤ 0
E=139:2:161

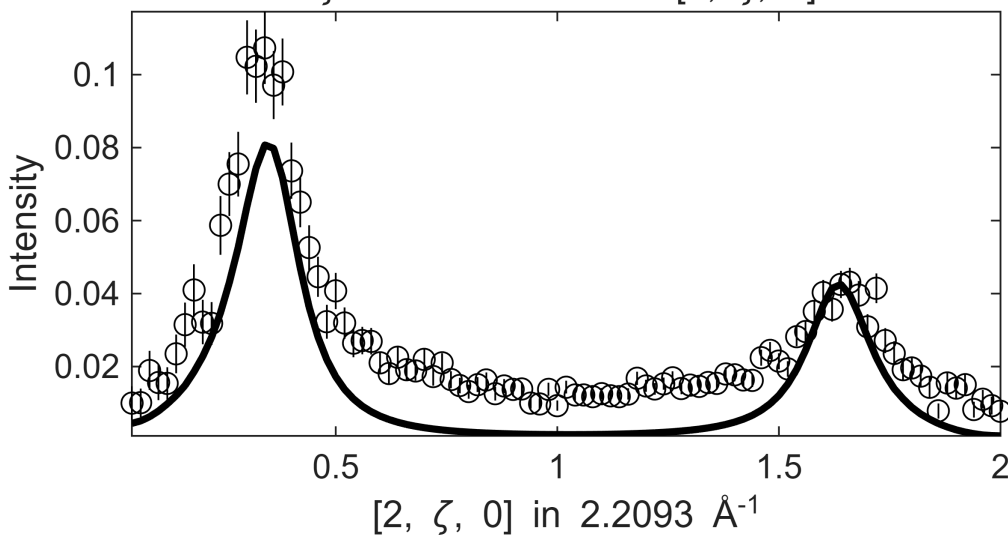


Fe_ei200_align_sym3.sqw

$S=0.67\pm0.21$; $J_0=35\pm 1.9$; $\gamma=76.0702\pm21.16$

$$\xi \leq 0.1 \text{ in } [2-\xi, 0, 0], -0.1 \leq \eta \leq 0.1 \text{ in } [2, 0, \eta], 140 \leq E$$
 $\zeta = -0.01:0.02:2.01$ in $[2, \zeta, 0]$ 

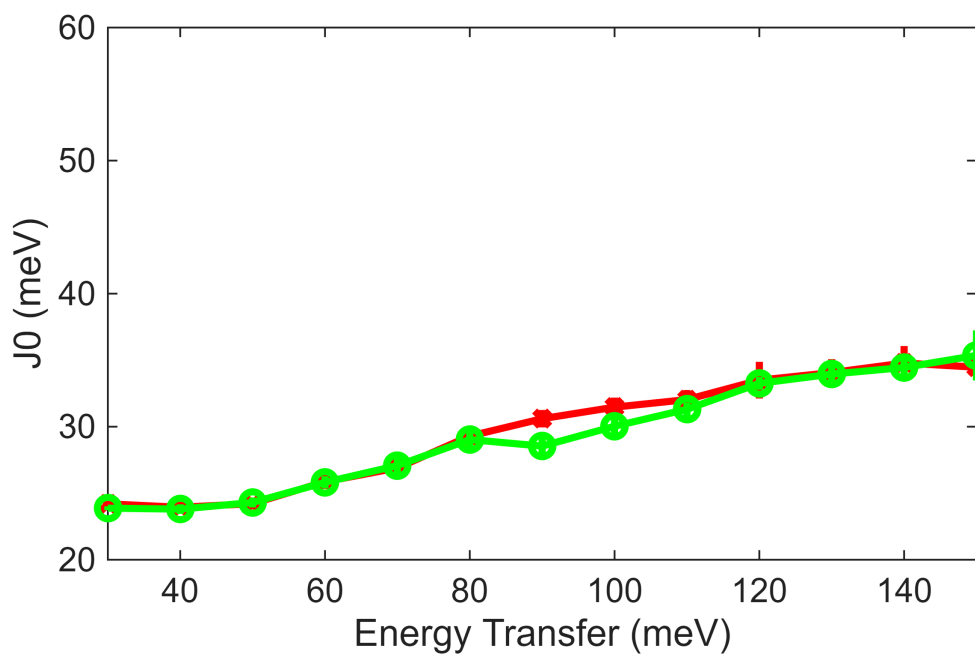
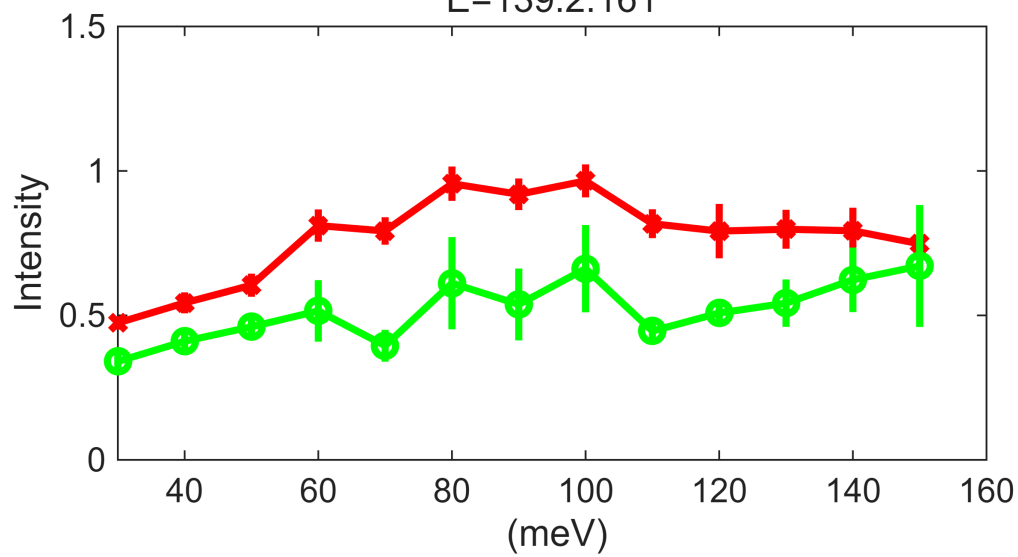
Fe_ei200_align_sym3.sqw

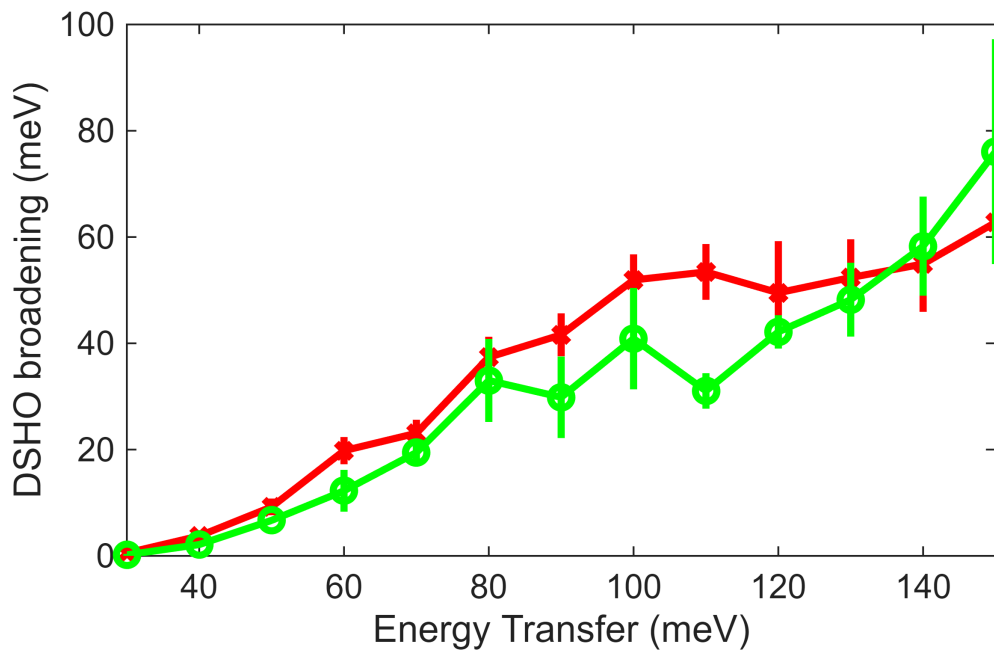
$$\xi \leq 0.1 \text{ in } [2-\xi, 0, 0], -0.1 \leq \eta \leq 0.1 \text{ in } [2, 0, \eta], 140 \leq E$$
 $\zeta = -0.01:0.02:2.01$ in $[2, \zeta, 0]$ 

[illegible]

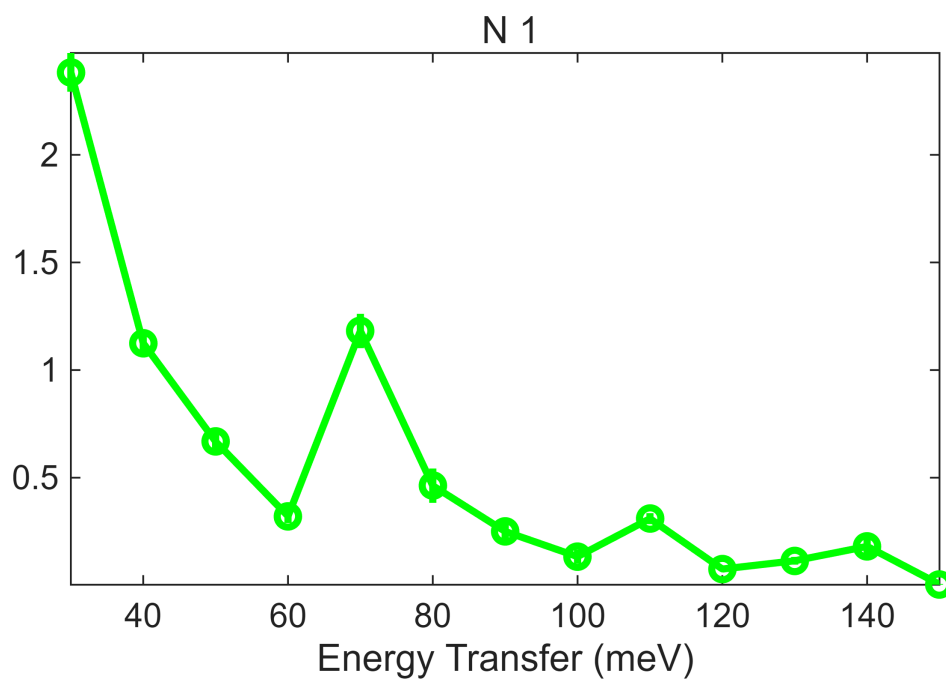
```
plot_SJG_results(fit_res, 'g', {'S', 'J0', 'gamma'}, {[0,1.5],[20,60],[0,100]},  
{Sg,J0,G});
```

Fe_ei200_align_sym3.sqw
 $\zeta = 1.8$ in $[2, \zeta, 0]$, $-0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$
 E=139:2:161

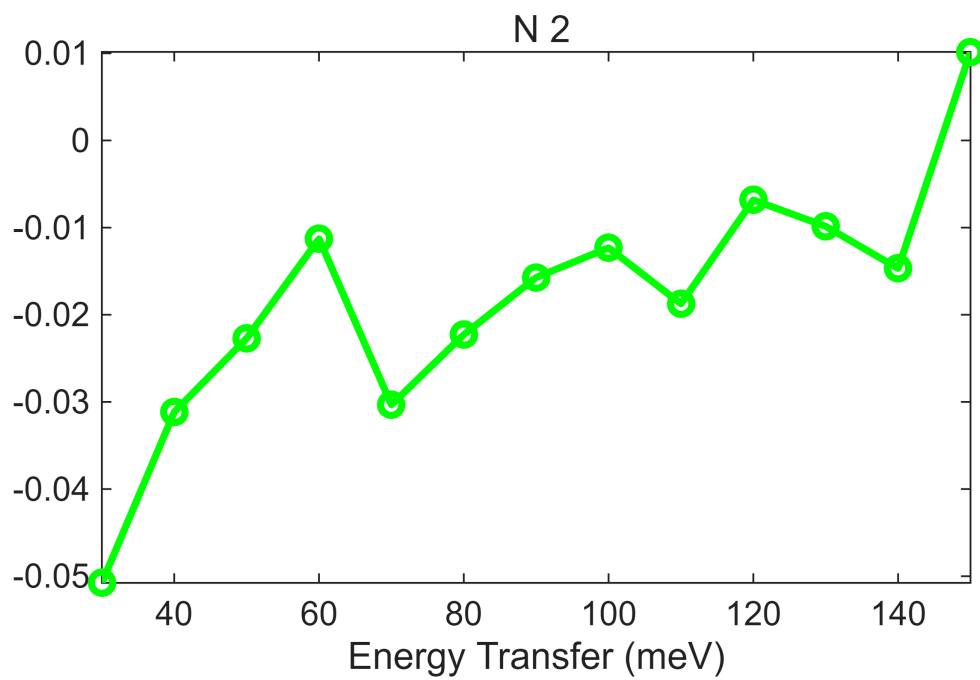




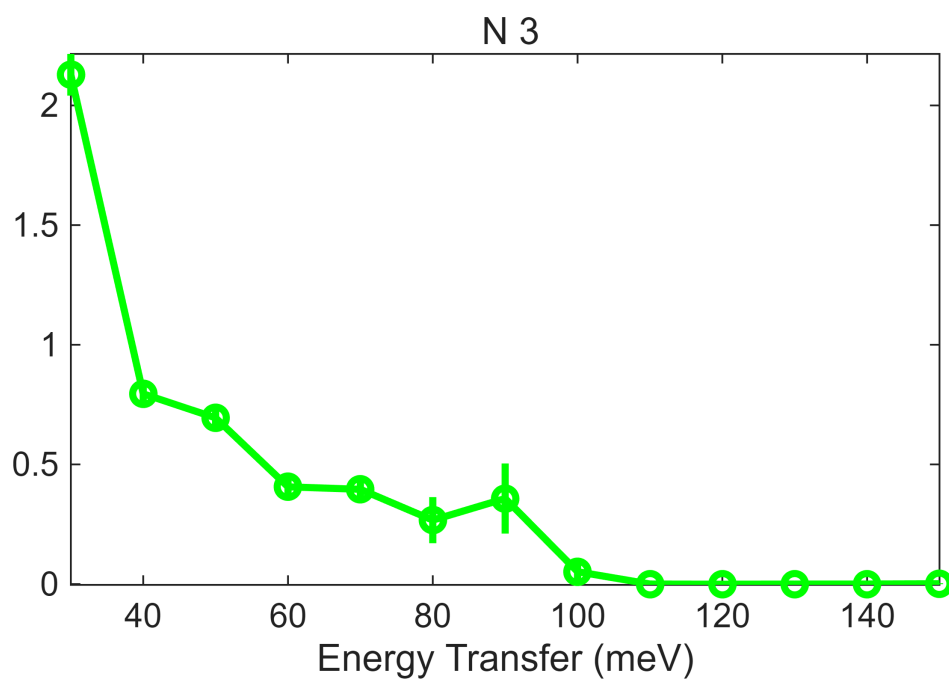
```
%bg_data = extract_bg_par(fit_res.peak_fit_par,3);
%pd(bg_data(1));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_
figure;%ly -0.05 0.05
bg_data = extract_bg_par(fit_res.peak_fit_par,[1,2]);
% original slope parameters
pd(bg_data(1));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_f
figure;%ly -0.05 0.05
```



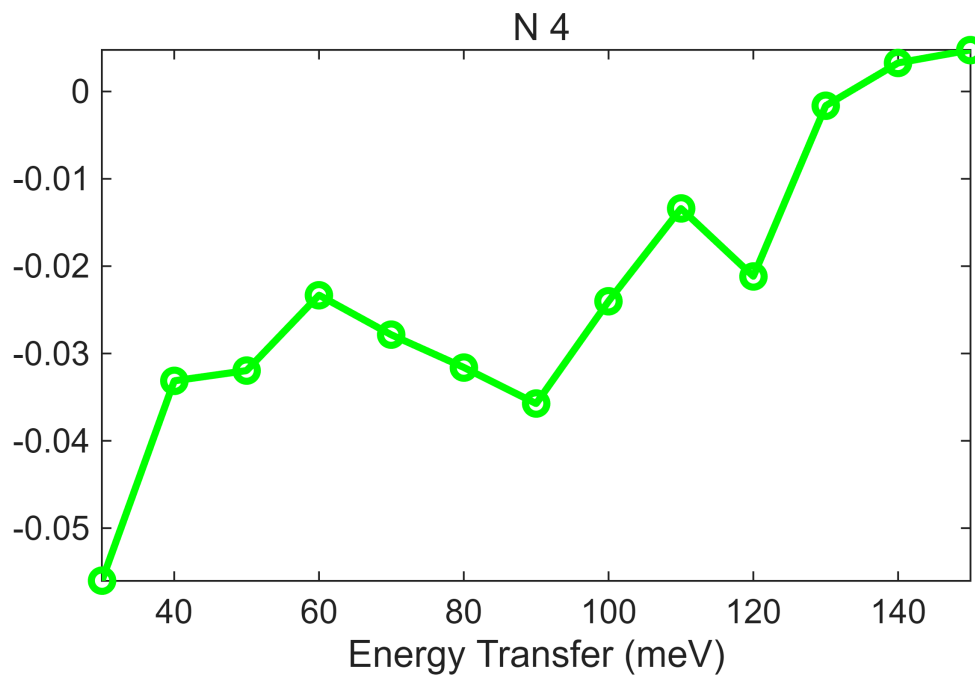
```
pd(bg_data(2));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_f
figure;%ly -0.05 0.05
```

```
pd(bg_data(3));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_f
figure;%ly -0.05 0.05
```



```
pd(bg_data(4));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_f
figure;%ly -0.05 0.05
```



```

fit_res.Speak_loc_sl2p = Sg;
fit_res.J0peak_loc_sl2p = J0;
fit_res.gamma_peak_loc_sl2p = G;

% fit_res.Speak = Sg;
% fit_res.J0peak = J0;
% fit_res.gamma_peak = G;

fit_res.peak_fit2Peaks_LocSlope2p_par = fit_res.peak_fit_par;
fit_resEi200 = fit_res;
save(fullfile(fit_data_path,fit_data_file),"fit_resEi200");

```